Import and Export Data

Day 1 - Introduction to Data Analysis with R

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The tidyverse

The tidyverse

> The tidyverse is an opinonated **collection of R packages** designed for data science. All packages share an underlying design philosophy, grammar, and data structures.
>
> (www.tidyverse.org)

These are the main packages from the tidyverse that we will use:

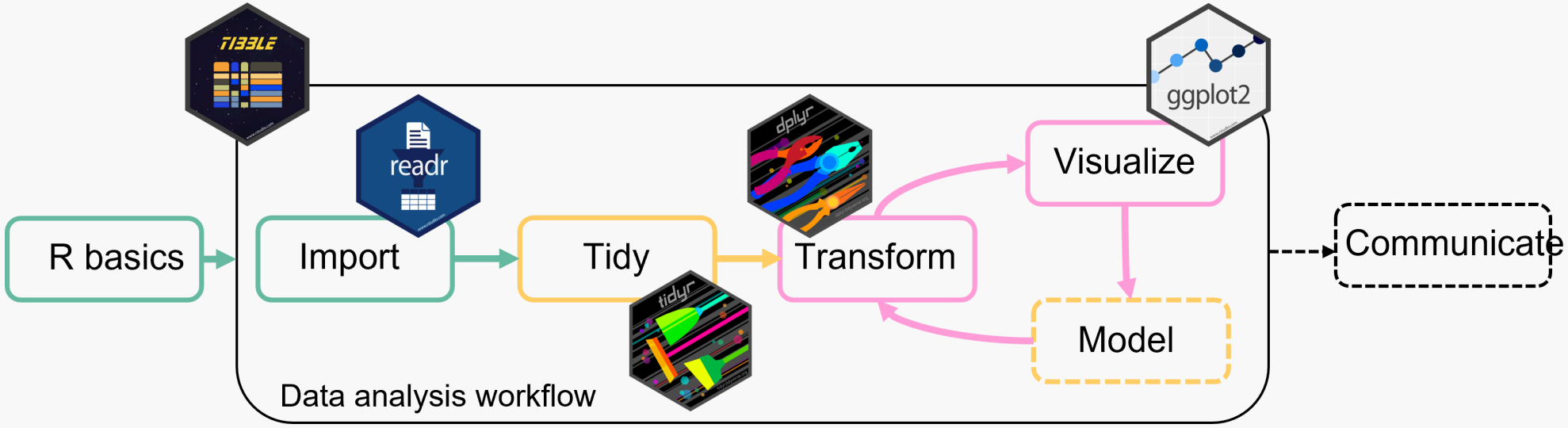


Image adapted from Wickham & Grolemund: R for Data Science

The tidyverse

Install the tidyverse once with:

```
install.packages("tidyverse")
```

Then load and attach the packages at the beginning of your script:

```
library(tidyverse)
```

You can also install and load the tidyverse packages individually, but since we will use so many of them together, it's easier to load and attach them together.

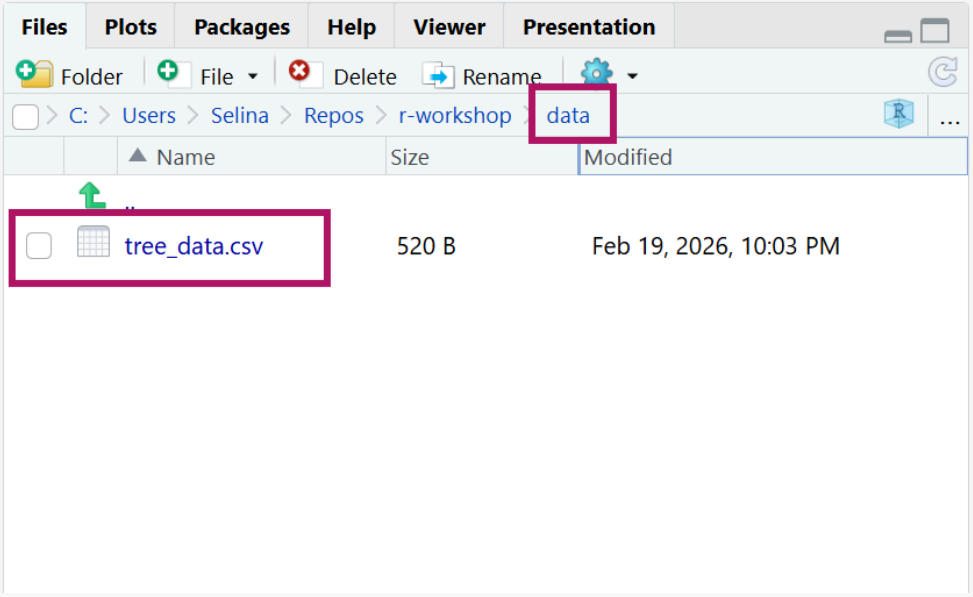
Import and export text data with readr

Reading CSV files with readr

```
library(tidyverse)
trees <- read_csv("data/tree_data.csv")
```

Where does R look for the file?

- "data/tree_data.csv" is a **relative path**: it starts from your project folder
- R looks for files starting from the project folder
- Your data files need to be inside your project for R to find them easily



Other `read_*` functions

All `read_*` functions take a path to the data file as a first argument:

`read_*("path/to/your/file", …)`

```
# semicolon delimiter
dat <- read_csv2("data/your_data.csv")
# tab delimiter
dat <- read_tsv("data/your_data.txt") # tab delimiter
# any delimiter, e.g. ";" or "----"
dat <- read_delim("data/your_data.csv", delim = ";")
dat <- read_delim("data/your_data.txt", delim = "----")
```

Dealing with messy files

The read functions provide several options for non-perfect data.

Have a look at `?read_delim` for all options.

Useful if your data is not a "perfect table"

Skipping lines

Use `skip` to skip lines at the top of a file

```r
# skip 4 metadata lines
read_csv(
  "data/tree_data_metadata_on_top.csv",
  skip = 4
)
#> # A tibble: 4 × 3
#>   tree_id   age circumference
#>     <dbl> <dbl>         <dbl>
#> 1       1     5            30
#> 2       2    10            60
#> 3       3    15            90
#> 4       4    20           120
```

```
Date of data collection: 01.01.2026
Name of collector: Selina
Temperature: 25°C

tree_id, age, circumference
1, 5, 30
2, 10, 60
3, 15, 90
4, 20, 120
```

Missing or bad headers

Use `col_names` if your file has no header row

```r
# assign default column names
read_csv(
  "data/tree_data_no_header.csv",
  col_names = FALSE
)
#> # A tibble: 4 × 3
#>      X1    X2    X3
#>   <dbl> <dbl> <dbl>
#> 1     1     5    30
#> 2     2    10    60
#> 3     3    15    90
#> 4     4    20   120
```

```
1, 5, 30
2, 10, 60
3, 15, 90
4, 20, 120
```

Missing or bad headers

Use `col_names` if your file has no header row

```
# specify custom column names
read_csv(
  "data/tree_data_no_header.csv",
  col_names = c("tree_id", "age", "circumference")
)
#> # A tibble: 4 × 3
#>   tree_id   age circumference
#>     <dbl> <dbl>         <dbl>
#> 1       1     5            30
#> 2       2    10            60
#> 3       3    15            90
#> 4       4    20           120
```

```
1, 5, 30
2, 10, 60
3, 15, 90
4, 20, 120
```

Write files with `write_*()`

Every `read_*` has a corresponding `write_*` function to export data from R.

```
# Comma delimiter
write_csv(dat, file = "data-clean/your_data.csv")
# Semicolon delimiter
write_csv2(dat, file = "data-clean/your_data.csv")
# tab delimiter
write_tsv(dat, file = "data-clean/your_data.txt")
# Any delimiter, e.g. ";" or "----"
write_delim(dat, file = "data-clean/your_data.csv", delim = ";")
write_delim(dat, file = "data-clean/your_data.txt", delim = "----")
```

Import excel files

Reading Excel files with readxl

The `readxl` package is part of the tidyverse, but you need to load it explicitly

```
library(readxl)
```

Use `read_excel()` to read an Excel file:

```
trees <- read_excel("data/tree_data.xlsx")
```

By default, this reads the first sheet. You can read other sheets with:

```
trees <- read_excel("data/tree_data.xlsx", sheet = "sheetName")
trees <- read_excel("data/tree_data.xlsx", sheet = 2)
```

- `read_excel` also has other functionality, like skipping rows etc.
- Check out `?read_excel` and the package documentation for more functionality

Reading Excel files with readxl

A little warning:

- Reading from a text file (.txt or .csv) is more reliable
- Be careful with complicated excel sheets with formulas etc.
- Always double check the data that you imported, e.g. by using the `summary` function and checking if the number of rows etc. is correct

Other read and write options

- Write Excel file with the writexl package (`write_excel`)
- Read and write SPSS, Stata, and SAS files with the haven package (`read_dta`, `read_spss`, `write_dta`, `write_spss`, …)
- Read and write Google Sheets with the googlesheets4 package (`read_sheet`, `write_sheet`)
- …

Important for all read_* functions

```
trees <- read_csv("data/tree_data.csv")
```

- Save the result of `read_*()` in a variable (e.g. `trees`) to work with it later
- Choose a meaningful name for the variable (e.g. `trees` instead of `data` or `my_data`)
- Use the function help `?` if in doubt
  - Shows you all available arguments and examples.

Now you

Task (25 min)

Read and write data files

Find the task description here

Guidelines for reading data sets in R

Use relative paths

```r
# Absolute path
trees <- read_csv(
  "C:/Users/Selina/Repos/Teaching/intro-r-data-analysis/data/tree_data.csv"
)
# Relative path
trees <- read_csv("data/tree_data.csv")
```

- Relative paths are interpreted relative to the **working directory**
- In RStudio projects, the **working directory is always the project root**

Why?

- Portable: will also work on other machines that copied the project
- More readable and less error prone

Tips for clean data import

- Prefer machine-readable file formats (`.csv`, `.txt` instead of `.xlsx`)
- No white space in column headers
  - Use a character as separator, e.g. `species_name` instead of `species name`
  - If this is unpractical, have a look at `janitor::clean_names()` from the `janitor` package
- No special characters in column headers (ä, ß, é, ê, %, °C, µ ...)
- Use `.` as a decimal separator (not `,`)
- Avoid white space and special characters in paths and file names
  - `data-raw/my_data.csv` instead of `data raw/my data.csv`

Summary I

Reading text files

```
# Comma separated (e.g. 1.5, 2.3, 3.1)
dat <- read_csv("data/your_data.csv")
# Semicolon separated (e.g. 1,5; 2,3; 3,1)
dat <- read_csv2("data/your_data.csv")
# Tab separated
dat <- read_tsv("data/your_data.txt")
# Any delimiter
dat <- read_delim("data/your_data.csv", delim = ";")
```

Writing text files

```
write_csv(dat, file = "data-clean/your_data.csv")
write_csv2(dat, file = "data-clean/your_data.csv")
write_tsv(dat, file = "data-clean/your_data.txt")
write_delim(dat, file = "data-clean/your_data.csv", delim = ";")
```

Summary II

Reading Excel files

```
library(readxl)
dat <- read_excel("data/your_data.xlsx")
dat <- read_excel("data/your_data.xlsx", sheet = "sheetName")
dat <- read_excel("data/your_data.xlsx", sheet = 2)
```

Dealing with messy files

```
# Skip metadata lines at the top
dat <- read_csv("data/your_data.csv", skip = 4)
# No header row — assign default names
dat <- read_csv("data/your_data.csv", col_names = FALSE)
# No header row — assign custom names
dat <- read_csv("data/your_data.csv", col_names = c("id", "age", "size"))
```